

Solution 07
Tracking MPC and LQR

Exercise 1 **MPC Tracking**

- 1) The problem is convex because the following conditions hold: convex cost function, linear/affine equality constraints, convex inequality constraint. Our problem has a convex cost function, as any norm is convex (epigraph reformulation of infinity norm possible).
- 2) We need the following conditions to hold:

$$\begin{aligned}x_s &= Ax_s + Bu_s \\ r &= Cx_s.\end{aligned}$$

Therefore,

$$\begin{aligned}x_s &= C^{-1}r, \\ x_s &= \begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}\end{aligned}$$

and with x_s given:

$$\begin{aligned}\begin{bmatrix} 1 \\ 3 \end{bmatrix} &= \begin{bmatrix} 7 & 2 \\ -2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + \begin{bmatrix} 6 \\ 2 \end{bmatrix} u_s \\ \begin{bmatrix} 1 \\ 3 \end{bmatrix} &= \begin{bmatrix} 13 \\ 7 \end{bmatrix} + \begin{bmatrix} 6 \\ 2 \end{bmatrix} u_s \\ u_s &= -2.\end{aligned}$$

The steady state x_s and input u_s can also be found directly using the problem introduced in the slides:

$$\begin{aligned}\min u_s^\top R u_s \\ \begin{bmatrix} I - A & -B \\ C & 0 \end{bmatrix} \begin{bmatrix} x_s \\ u_s \end{bmatrix} &= \begin{bmatrix} 0 \\ r \end{bmatrix} \\ x_s &\in \mathcal{X} \quad u_s \in \mathcal{U}.\end{aligned}$$

- 3) Define $\Delta x = x - x_s$, $\Delta u = u - u_s$ and insert x , u into the original optimization problem:

$$\begin{aligned}\min_U \quad & P \|r - C(\Delta x_2 + x_s)\|_\infty + \sum_{i=0}^1 Q \|r - C(\Delta x_i + x_s)\|_\infty + R (u_s - (\Delta u_i + u_s))^2 \\ \text{s.t.} \quad & x_0 = x(0), \\ & (\Delta x_{i+1} + x_s) = A(\Delta x_i + x_s) + B(\Delta u_i + u_s), \quad i = 0, 1 \\ & u_{\min} \leq (\Delta u_i + u_s) \leq u_{\max}, \quad i = 0, 1,\end{aligned}$$

which leads to

$$\begin{aligned} \min_U \quad & P \|C\Delta x_2\|_\infty + \sum_{i=0}^1 Q \|C\Delta x_i\|_\infty + R (\Delta u_i)^2 \\ \text{s.t.} \quad & x_0 = x(0), \\ & \Delta x_{i+1} = A\Delta x_i + B\Delta u_i, \quad i = 0, 1 \\ & u_{\min} - u_s \leq \Delta u_i \leq u_{\max} - u_s, \quad i = 0, 1, \end{aligned}$$

where we used

$$\begin{aligned} x_s &= Ax_s + Bu_s \\ r &= Cx_s. \end{aligned}$$

Exercise 2 Tracking LQR

1) (i) Steady-state target expression

$$\begin{bmatrix} A - I & B \\ C & 0 \end{bmatrix} \begin{bmatrix} x \\ u \end{bmatrix} = \begin{bmatrix} 0 \\ r \end{bmatrix}$$

(ii) This is exactly the same as LQR regulation

$$\begin{aligned} \min \quad & \sum_{i=0}^{\infty} \Delta x_i^T Q \Delta x_i + \Delta u_i^T R \Delta u_i \\ \text{s.t.} \quad & \Delta x_{i+1} = A\Delta x_i + B\Delta u_i \end{aligned}$$

where $\Delta x = x - x_s$ and $\Delta u = u - u_s$.

(iii) If P is a solution to the DARE

$$P = Q + A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A$$

then the control law is

$$u(x, r) = K(x - x_s) + u_s$$

where $K = -(R + B^T P B)^{-1} B^T P A$

2) Target selection problem

$$\begin{aligned} x_s &= 2x_s + u_s & r &= x_s \\ u_s &= -r & r &= x_s \end{aligned}$$

DARE

$$\begin{aligned} P &= Q + A^T P A - A^T P B (R + B^T P B)^{-1} B^T P A \\ P &= 1 + 4P - 2P(1 + P)^{-1}2P \\ P &= 2 + \sqrt{5} \end{aligned}$$

LQR

$$\begin{aligned} K &= \frac{-2P}{1 + P} \\ &= -\frac{4 + 2\sqrt{5}}{3 + \sqrt{5}} \end{aligned}$$

Controller

$$u(x, r) = K(x - x_s) + u_s = -\frac{4 + 2\sqrt{5}}{3 + \sqrt{5}}(x - r) - r$$